Module-VII
Lecture-I
Introduction to Digital VLSI Testing

1. Introduction to Philosophy of Testing

"If anything can go wrong, it will"--A very well known statement known as Murphy's Law. So, to ensure that only fault free systems are delivered, before deploying any system in the field or shipping a product to a customer, it needs to be *Tested.* Testing a system comprises subjecting it to inputs and checking its outputs to verify whether it behaves as per the specifications targeted during design. Let us take the simple example of an electric iron. From our daily life experience, testing in this case would be to plug it in 220V AC and see if is heating. In this simple example also, testing may not be so straight foreword as mentioned above. Test for heating is just verifying its "functional" specification, *that also partially*. A near complete test will require the following

- Safety:
    - All exposed metal parts of the iron are grounded
    - Auto-off on overheating
- Detailed Functionality
    - Heating when powered ON.
    - Glowing of LED to indicate power ON.
    - Temperature matching with specification for different ranges that can be set using the regulator (e.g., woolen, silk, cotton etc.)
- Performance
    - Power consumption as per the specifications
    - Time required to reach the desired temperature when range is changed using the regulator

The story is not over. The list discussed above is the set of tests for ONLY *electrical parameters*. Similarly, there will be a full list of tests for *mechanical parameters*, like maximum height from which there is resistance to breaking of plastic parts if dropped on a tiled floor etc.

The number of tests performed depends on the time being allocated and which in turn is decided by the target price of the product. Increasing the number of tests lead to more test time, requirement of sophisticated equipments, working hours of experts etc. thereby adding to cost of the product. In case of example of the iron, if it is a cheap one (may be local made) the only test is to verify "if it heats", while if it is branded, a lot more tests are done to assure quality, safety and performance of the product.

Now let us take an example of testing a NAND gate, shown in Figure 1.

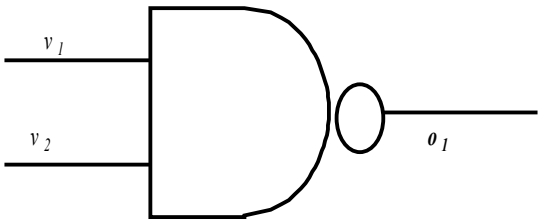


Figure 1. NAND gate

The simple test to verify the proper functionality of the NAND gate would comprise subjecting the gate with inputs listed in Table 1 and then checking if the outputs match the ones listed in the table.

Table 1. Test set for NAND gate

Input		**Output**
v_1	v_2	o_1
0	0	1
0	1	1
1	0	1
1	1	0

Just like the example of the electric iron, this test for the NAND gate is just the starting point. A more detailed test can be enumerated as follows

Detailed tests for the NAND gate

- Digital Functionality
    - Verify input/output of Table 1
- Delay Test
    - 0 to 1: time taken by the gate to rise from 0 to 1.
        - *v1=1, v2=1* changed to *v1=1, v2=0;* After this change in input, time taken by o_1 to change from 0 to 1.
        - *v1=1, v2=1* changed to *v1=0, v2=1;* After this change in input, time taken by o_1 to change from 0 to 1.
        - *v1=1, v2=1* changed to *v1=0, v2=0;* After this change in input, time taken by o_1 to change from 0 to 1.
    - 1 to 0: time taken by the gate to fall from 1 to 0.
        - *v1=0, v2=0* changed to *v1=1, v2=1;* After this change in input, time taken by o_1 to change from 1 to 0.
        - *v1=1, v2=0* changed to *v1=1, v2=1;* After this change in input, time taken by o_1 to change from 1 to 0.
        - *v1=0, v2=1* changed to *v1=1, v2=1;* After this change in input, time taken by o_1 to change from 1 to 0.
- Fan-out capability:
    - Number of gates connected at o_1 which can be driven by the NAND gate.
- Power consumption of the gate
    - Static power: measurement of power when the output of the gate is not switching. This power is consumed because of leakage current
    - Dynamic power: measurement of power when the output of the gate switches from 0 to 1 and from 1 to 0.
- Threshold Level
    - Minimum voltage at input considered at logic 1
    - Maximum voltage at input considered at logic 0
    - Voltage at output for logic 1
    - Voltage at output for logic 0
- Switching noise

    - Noise generated when the NAND gate switches from 0 to 1 and from 1 to 0
- Test at extreme conditions
    - Performing the tests at temperatures (Low and High Extremes) as claimed in the specification document.

These tests are for the “logic level [1,2]” implementation of the NAND gate.

Figure 2 shows the detailed CMOS level implementation of the gate. Tests for the NAND gate at transistor level would be many fold more than the ones discussed above [3]. They would comprise verifying all parameters of a transistor. Some tests for the NMOS (T3) are enumerated below,

- Output Characteristics
    - a set of I_{DS} vs V_{DS} curves for different constant values of the gate-source voltage V_{GS}
- Transfer characteristics
    - a set of I_{DS} vs V_{GS} curves for different values of the substrate-source voltage V_{BS}, at constant V_{DS}
- Threshold Voltage Test
    - Threshold Voltage obtained in test, matches the specifications

Further, all the tests for a transistor need to be repeated for all the four transistors involved. Last but not the least, Transistor T1 and T2 are to be matched (i.e., equivalent in terms of parameters) and similarly for T3 and T4. To summarize, as more details of implementation of a gate (or any circuit) is considered, test procedure becomes more complex.

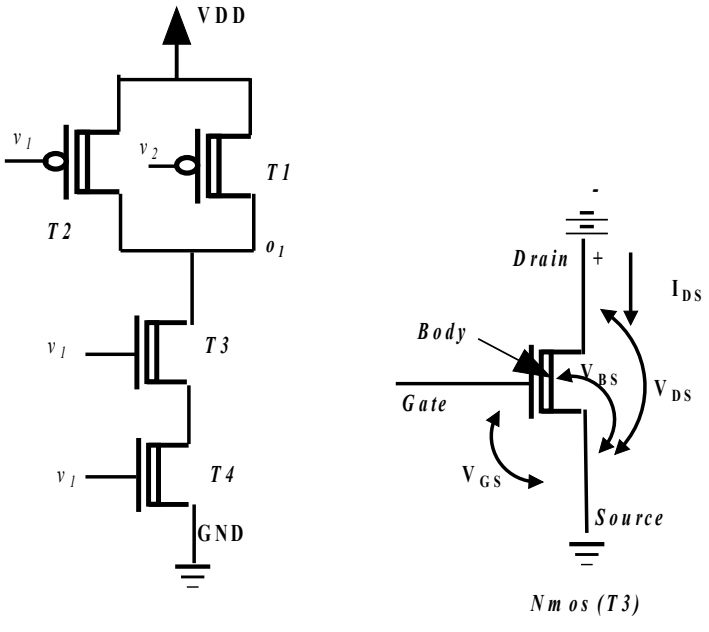


Figure 2. CMOS implementation of the NAND gate

Now, Figure 3 shows the layout of the gate in a silicon die. Once can easily estimate the number of tests by looking at the increase in complexity from logic gate level details to silicon level details.

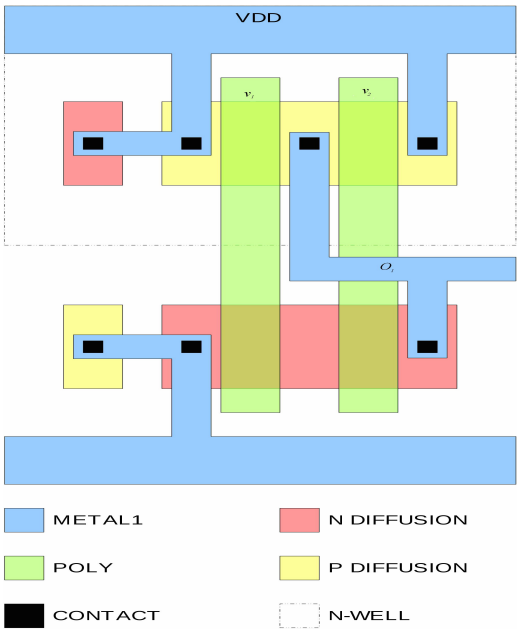


Figure 3. Layout of a NAND gate in a silicon die

So now the question is, *"given a digital logic gate, what tests are to be performed to assure an acceptable quality of product at reasonable price*".

Ideally speaking, all testes at silicon level, transistor level and logic level are to be performed. It is to be noted that in a typical digital IC, there are some tens of thousands of logic gates and about a million samples to be tested. So time for complete testing of the ICs would run into years. Thus, test set for the NAND gate should be such that results are accurate (say 99% above) yet time for testing is low (less than a millisecond). Under this requirement, it has been seen from experience over a large number of digital chips that nothing more than validating the logical functionality (Table 1 for the NAND gate) and at proper time (i.e., timing test) can be accomplished. Latter we will see that not even the full logic functionality of a typical circuit can be tested within practical time limits.

Now we define DIGITAL TESTING. DIGITAL TESTING is *not* testing digital circuits (comprised of logic gates); as discussed, all tests possible for a digital circuit are not applied in practical cases. ***DIGITAL TESTING is defined as testing a digital circuit to verify that it performs the specified logic functions and in proper time.***

For the discussion so far, it may appear that testing digital VLSI circuits and other systems like electrical iron are similar. However, there are some fundamental differences that make VLSI circuit testing more important step to assure quality, compared to classical systems like electric iron, fan etc.

- The intension to make single chip implementation of complex systems has reached a point where effort is made to put millions of transistors on a single chip and increase the operation speed to more than a GHz. This has initiated a race to move into deep sub-micron technology, which also increases the possibility of faults in the fabricated devices. Just when a technology matures and faults tend to decrease, a new technology based on lower sub-micron devices evolves, thereby always keeping testing issues dominant. Figure 4 shows the transistor count (which rises with lowering the sub-micron of the manufacturing technology) versus years. This is unlike traditional systems where the basic technology is matured and well tested, thereby resulting in very less number of faults.
- In case of detection of faults in a traditional system it is diagnosed and repaired. However, in case of circuits, on detection of a fault the chip is binned as defective and scrapped (i.e., not repaired). In other words, in VLSI testing chips are to be binned as normal/faulty so that only fault free chips are shipped and no repairing is required for faulty ones.

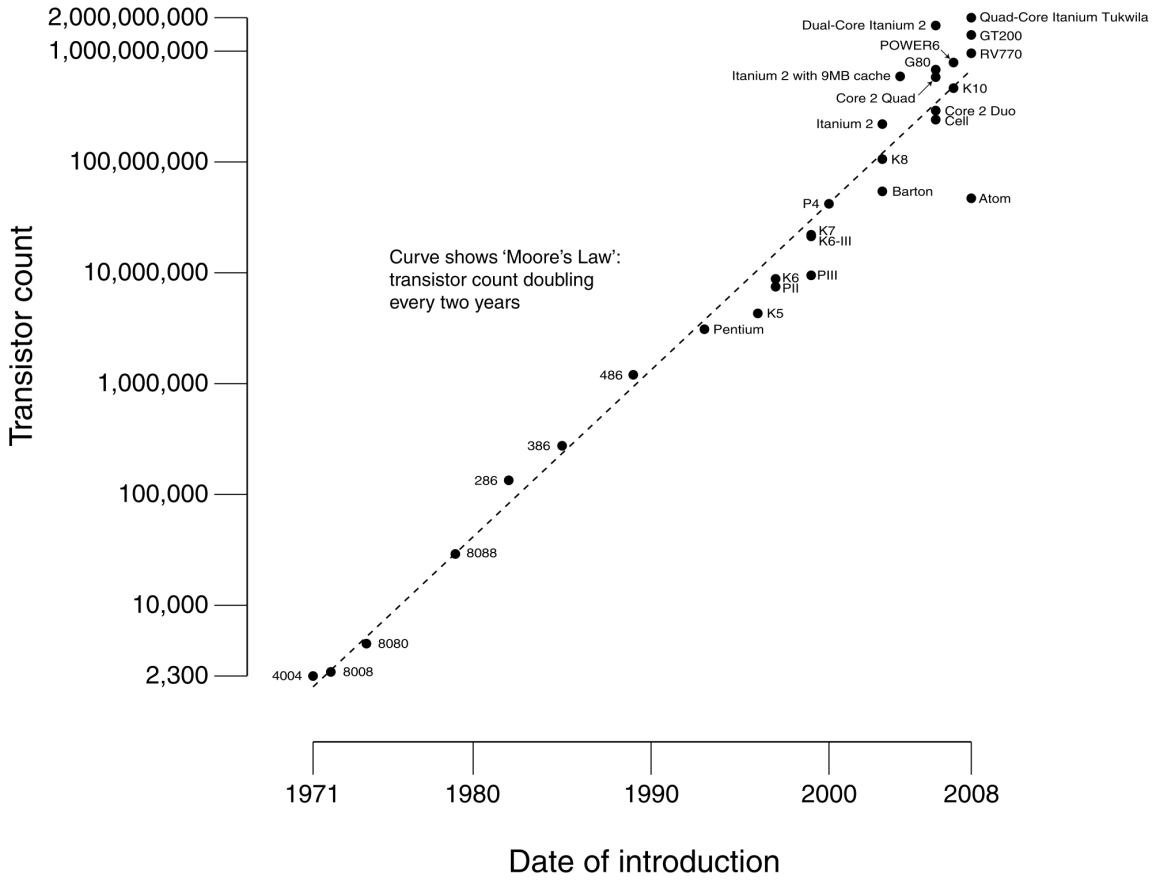


Figure 4. Transistor count versus years (taken from [5])

2. Digital Testing In A VLSI Design Flow

Figure 5 illustrates a typical digital VLSI design and test flow. This starts with the development of the specifications for the system from the set of requirements, which includes functional (i.e., input-output) characteristics, operating characteristics (i.e., power, frequency, noise, etc.), physical and environmental characteristics (i.e., packaging, humidity, temperature, etc.) and other constraints like area, pin count, etc. This is followed by an architectural design to produce a system level structure of realizable blocks for the functional specifications. These blocks are then implemented at resister transfer level (RTL) using some hardware definition language like Verilog or VHDL. The next step, called logic design, further decomposes the blocks into gates maintaining operating characteristics and other constraints like area, pin count, etc. Finally, the gates are implemented as physical devices and a chip layout is produced during the physical design. The physical layout is converted into photo masks that are used in the fabrication process. Fabrication consists of processing silicon wafers through a series of steps involving photo resist, exposure through masks, etching, ion implantation, etc. Backtrack from an intermediary stage of the design and test flow may be required if the design constraints are not satisfied. It is unlikely that all fabricated chips would satisfy the desired specifications. Impurities and defects in materials, equipment malfunctions, etc. are some causes leading to the mismatches. The role of testing is to detect the mismatches, if any. As shown in Figure 5, the test phase starts in form of planning even before the logic synthesis and hardware based testing is performed after the fabrication. Depending on the type of circuit and the nature of testing required, some additional circuitry, pin out, etc. need to be added with the original circuit so that hardware testing becomes efficient in terms of fault coverage, test time, etc.; this is called design for testability (DFT). After logic synthesis, (binary) test patterns are generated that need to be applied to the circuit after it gets manufactured. Also, the expected responses (golden response) for these test patterns are computed which are matched with the response obtained from the manufactured circuit.

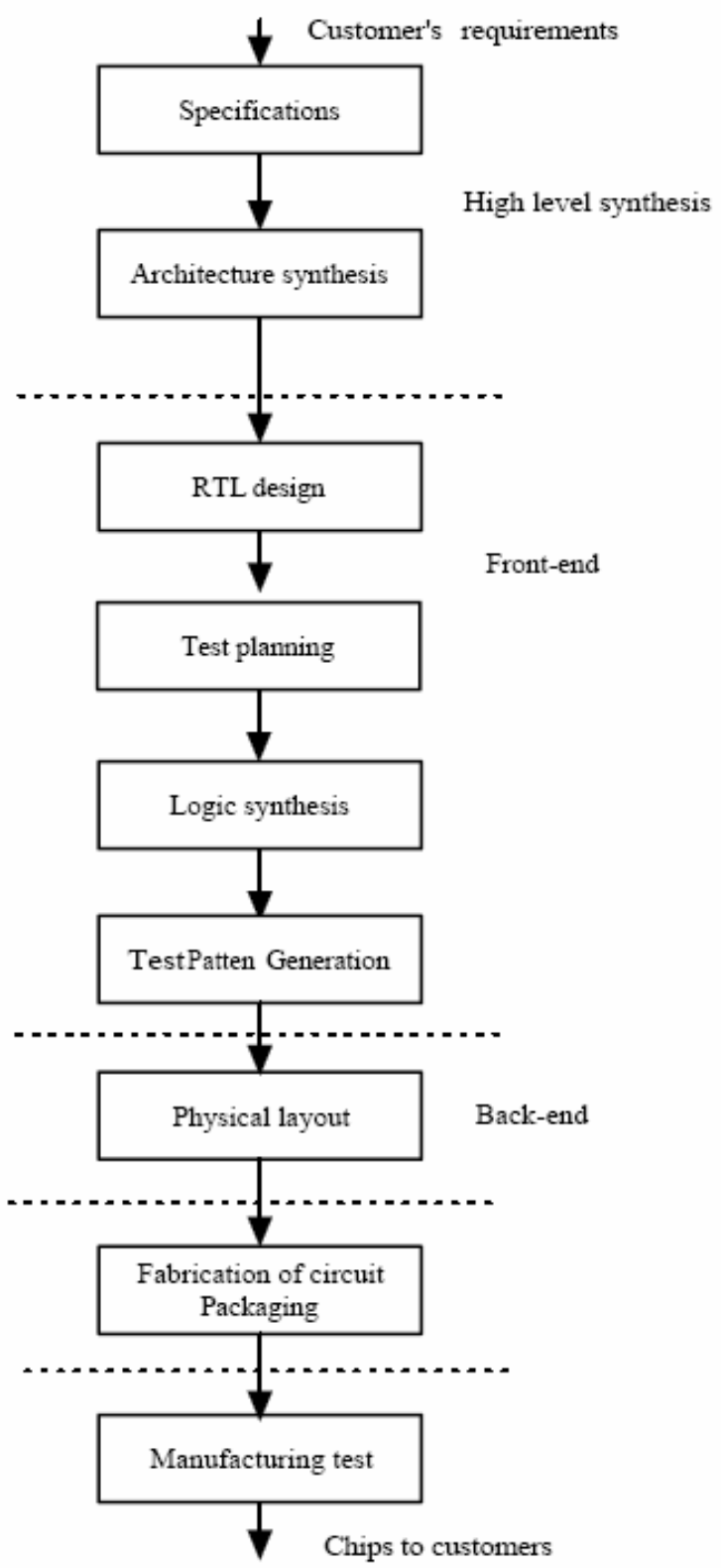


Figure 5. A typical digital VLSI design flow

Figure 6 illustrates the basic operations of digital testing on a manufactured circuit. Binary test vectors are applied as inputs to the circuit and the responses are compared with the golden signature, which is the ideal response from a fault-free circuit.

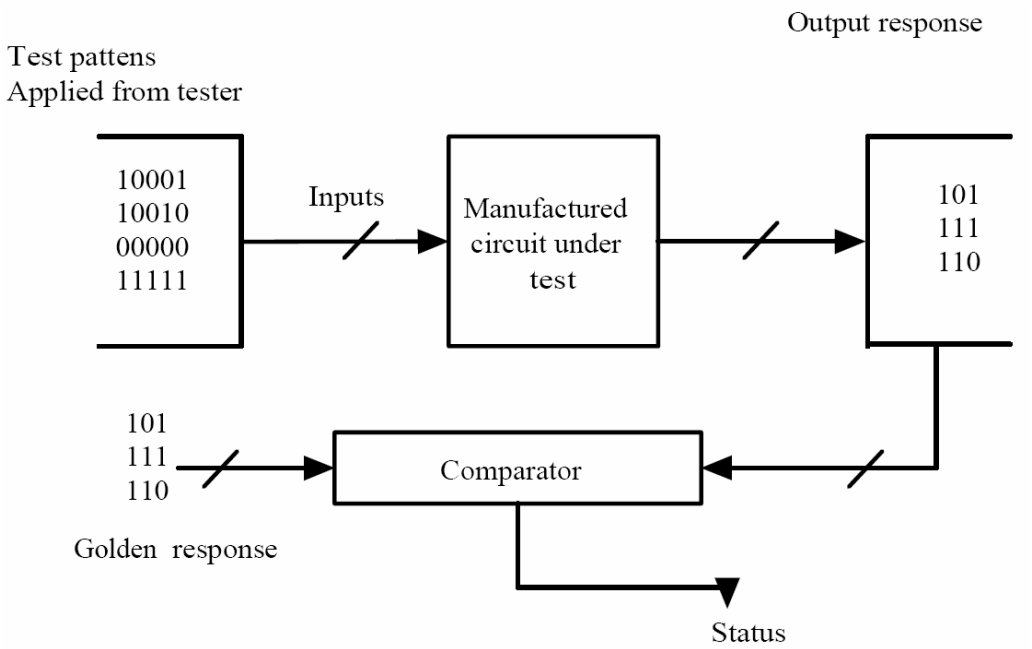


Figure 6. Digital VLSI test process

These test patters are generally applied and analyzed using automatic test equipment (ATE). Figure 7 shows the picture of an ATE from Teradyne [4].



Figure 7. Automatic Test Equipment

3. **Taxonomy of Digital Testing**

Digital testing can be classified according to several criteria. Table 2 summarizes the most important attributes of various digital testing methods and the associated terminology.

Table 2. Taxonomy of digital testing

Criterion	Attributes of testing method	Terminology
When tested?	1. Once after manufacture 2. Once before startup of circuit 3. Always during the system operation	1. Manufacturing Test 2. Built in self test (BIST) 3. On-line testing (OLT)
Where is the source of Test patterns?	1. An external tester 2. Within the chip 3. No patters applied, only monitoring	1. Automatic Test Equipment (ATE) based testing 2. BIST 3. OLT
Circuit in which form is being tested?	1. Wafer 2. IC 3. Board 4. System	1. Non packaged IC level testing 2. Packaged level testing 3. Board level testing 4. System level testing
How are the test patterns applied?	1. In a fixed predetermined order 2. Depending on results	1. Static Testing 2. Adaptive testing
How fast are the test patterns applied?	1. Much slower than the normal speed of operation 2. At normal speed of operation	1. DC (static) testing 2. At-speed testing
Who verifies the test results by matching with golden response?	1. On chip circuit 2. ATE	1. BIST 2. Automatic Test Equipment (ATE) based testing

4. Test Economics

The basic essence of economics of a product is "minimum investments and maximum returns". To under test economics the investments (price paid) and returns (gains) for a VLSI testing process are to be enumerated.

- Investments
  1. *Man hours for test plan development:*

     Expert test engineers are required to make elaborate test plans.
  2. *CAD tools for Automatic Test Pattern Generation*

     Given a circuit, binary input patters required for testing is automatically generated by commercial CAD tools.
  3. *Cost of ATE*

     ATE is a multimillion dollar instrument. So cost of testing a chip in an ATE is dependent on
     - time a chip is tested,
     - the number of inputs/outputs pins
     - frequency the test patters are to be applied
  4. DFT or BIST circuitry

     Additional circuitry kept on-chip to help in testing results in raise in chip area, thereby by increasing the unit price (because of more use of silicon). Further, power consumption of the chip may also rise due to addition of extra circuits. Also, it may be noted that if individual chips are larger in area, then more number of fabricated chips are found faulty (i.e., yield is less). To cope up with fewer yields, cost of individual unit is increased.

     At-speed testing is more effective than lowered-speed testing. However, the ATEs that can perform at-speed testing for the latest chips are extremely expensive. It may be noted that generating and capturing high speed patterns using on-chip circuitry is manifold simple and cheaper than doing so using an external tester (ATE). So additional DFT/BIST circuit is required to apply patterns and capture response at high speed. Once the response is captured, they can be downloaded by the ATE at a lower speed for analysis. The BIST/DFT circuitry will reduce the

yield of the VLSI chip, thus increasing the cost. As this cost increase is offset by cost reduction in ATE, this additional BIST/DFT circuitry is economically beneficial.

- Returns
    a) Proper binning of Chips:

       The more a testing process can be made perfect, the less will be the errors in binning the normal chips and the faulty ones. In case of VLSI testing, it is not of much concern as how many chips are binned as faulty, rather important is how many faulty chips are binned as normal. Faulty chips binned as normal are shipped and that compromises the quality of test solution and the brand name of the company. So, economic return from “VLSI testing” is the accuracy in shipping functionally perfect chips.

From the next lectures, we will go into the details of Digital VLSI Testing. The tentative breakup of the lectures would be the following

Module VII: Introduction to Digital Testing

Lecture-I: Introduction to Digital VLSI Testing
Lecture-II: Functional and Structural Testing
Lecture-III: Fault Equivalence

Module VIII: Fault Simulation and Testability Measures

Lecture-I, II and III: Fault Simulation
Lecture-IV: Testability Measures (SCOAP)

Module IX: Combinational Circuit Test Pattern Generation

Lecture-I: Introduction to Automatic Test Pattern Generation (ATPG) and ATPG Algebras
Lecture-II and III: D-Algorithm

Module X: Sequential Circuit Testing and Scan Chains

Lecture-I: ATPG for Synchronous Sequential Circuits
Lecture-II and III: Scan Chain based Sequential Circuit Testing

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Module XI: Built in Self test (BIST)

Lecture I and II: Built in Self Test
Lecture III and IV: Memory Testing

References

[1] J. A. Abraham and H.C. Shih. Transistor-level test generation for physical failures in CMOS circuits. Design Automation Conference, pages 243–249, 1986.

[2] M. Abramovici, M.A. Breuer, and A.D. Friedman. Digital Systems Testing and Testable Design. Wiley-IEEE Press, 1994.

[3] Phillip E. Allen, Douglas R. Holberg. CMOS Analog Circuit Design, Oxford University Press, Second Edition, 2002.

[4] http://www.teradyne.com/

[5] Moore's Law, From Wikipedia, Whttp://en.wikipedia.org/wiki/Moore%27s_law